Progressive exercises

E1 · Practice. In a basis of space, the vectors $\vec{u}$ and $\vec{v}$ have coordinates $(1, 2, -1)$ and $(3, 6, -3)$ respectively. Are they collinear? Justify.

E2 · Practice. In an orthonormal frame of space, consider $A(0, 1, 2)$ and $B(4, -1, 0)$. Determine the coordinates of the midpoint of $[AB]$, then calculate the distance AB.

E3 · Practice. In a basis of space, consider $\vec{u} = (1, 0, 1)$, $\vec{v} = (0, 1, 1)$ and $\vec{w} = (2, 3, 5)$. Express $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$, then deduce that these three vectors are coplanar.

E4 · Practice. In a basis of space, the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ have coordinates $(1, 0, 1)$, $(0, 1, 1)$ and $(1, 1, 0)$. Is the triple $(\vec{u}, \vec{v}, \vec{w})$ a basis of space? Justify.

E5 · Practice. In a frame of space, consider $A(1, 0, 2)$, $B(3, -1, 4)$ and $C(5, -2, 6)$. Determine whether C belongs to the line (AB). Justify with a vector calculation.

Textbook or past-paper work

Barbazo, p. 328, exercise 21. In class after S15: do all three questions. For each intersection, begin by identifying each of the planes from the vertices of the cube and the midpoints given. Distinguish between intersecting, coincident and strictly parallel planes.